

Cahier des charges PFA + projet Frontend

Ajouter l'aspect Entrepreneuriat et Gestion de projet

Projet : Application de gestion du pointage numérique des employés

1. Contexte et objectif

L'objectif est de développer une application web permettant de :

- Gérer le **pointage des employés en temps réel**
- Vérifier la **présence via géolocalisation**
- Scanner un **QR code sécurisé**
- Gérer les **congés**
- Offrir un **tableau de bord analytique**
- Détecter :
 - les **manquements (retards, absences)**
 - les **insuffisances d'heures travaillées**

2. Architecture technique

Frontend

- Framework : React ou Next js
- Librairies possibles :
 - Axios / Fetch
 - React Router
 - Chart.js / Recharts
 - Bootstrap / Material UI

Backend (au choix)

- Option 1 : Spring Boot
- Option 2 : Node.js Express js ou Next JS
- Option 3 : PHP (Laravel recommandé)

Base de données

- MongoDB

API

- REST API (JSON)

3. Acteurs du système

-  Administrateur
-  Employé

4. Fonctionnalités principales

4.1 Authentification

- Connexion sécurisée (email + mot de passe)
- Gestion des rôles (admin / employé)
- Session / JWT

4.2 Pointage intelligent

Fonctionnement :

1. L'utilisateur clique sur "Pointer"
2. L'application :
 - récupère la **géolocalisation GPS**
 - vérifie si l'utilisateur est dans la zone autorisée
3. Scan du **QR Code**
4. Enregistrement du pointage

Données enregistrées :

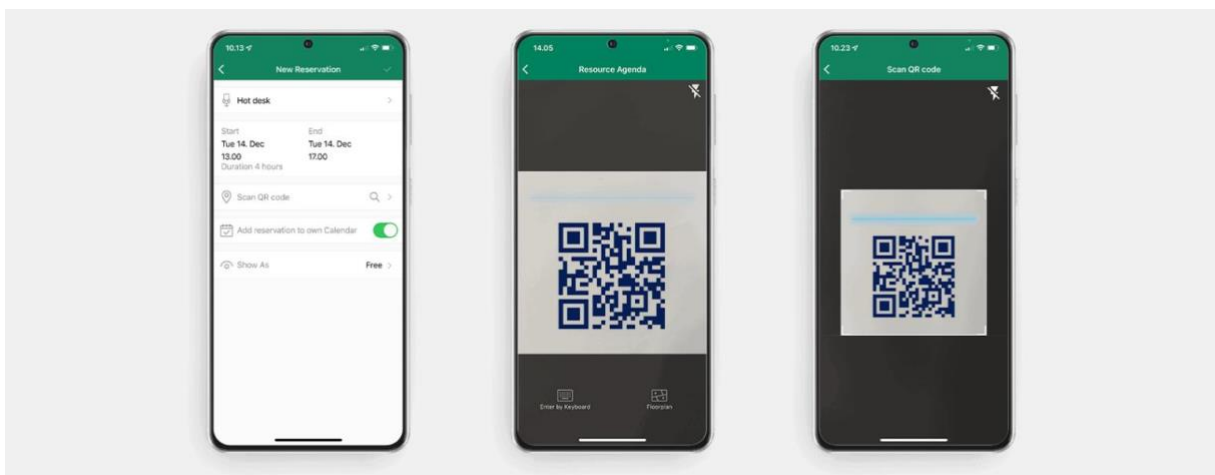
- utilisateur
- date / heure
- localisation (latitude, longitude)
- type : entrée / sortie

QR Code (Sécurité)



QR Code Attendance

Simply print the QR code, and employees can scan it to mark their attendance.

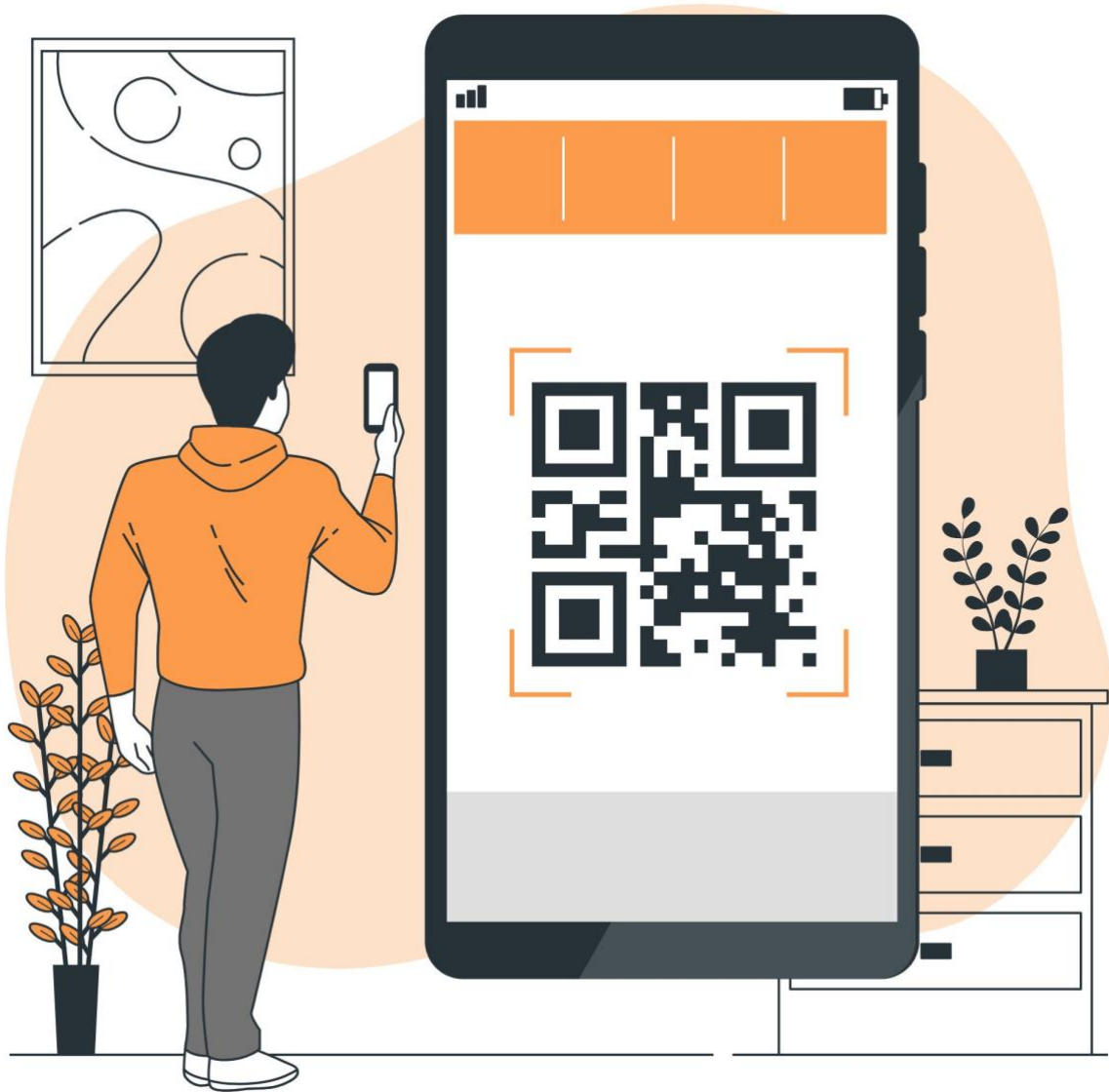


QR based time clocking automatic attendance system for field employees

This slide shows automatic attendance system for field employee based on qr code which allows to track real time information. It further includes details such as discover qr code ,scan qr code ,access information etc.



This slide is 100% editable. Adapt it to your needs and capture your audience's attention.







1



Registration with the visitors' personal information.

2



URL received upon entry of data to the system.

3



Visitors click on the links to obtain dynamic QR codes or barcodes.

4



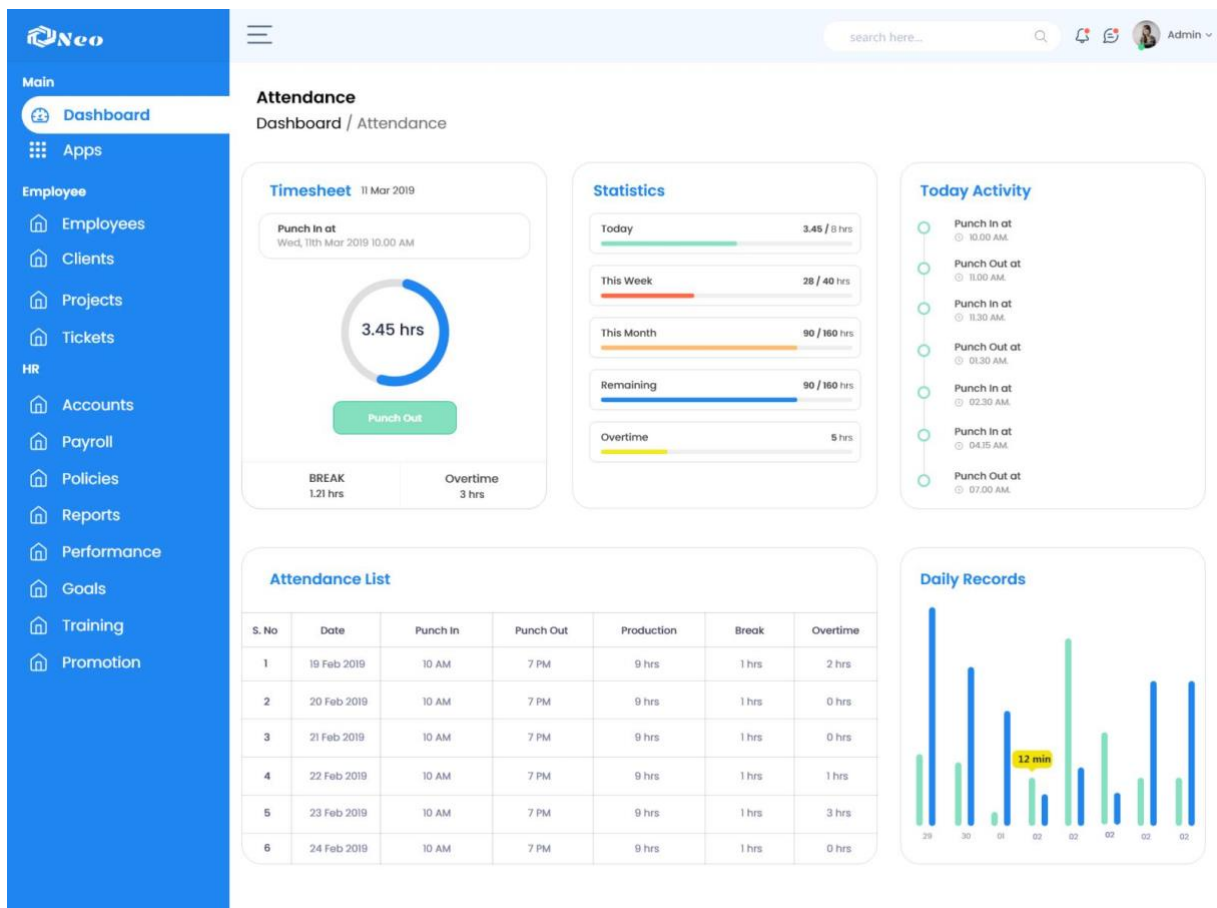
Within 30 seconds, registered visitors will be granted access with the same QR Code. Apart from building entrance, entrance control, car-park gates and elevators also support QR code access function.

- QR code généré par l'entreprise
- Changeable (rotation possible)
- Empêche le pointage frauduleux

4.3 Gestion des congés

- Demande de congé (employé)
- Validation / refus (admin)
- Types :
 - Congé annuel
 - Maladie
 - Exceptionnel

4.4 Tableau de bord



Select Date ⓘ

12/1/2021 - 5/31/2022

Select Employee ⓘ

All

Total Employees ⓘ

28

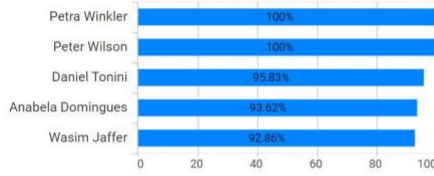
Absences Days ⓘ

102

Unapproved Leave ⓘ

4

Top 5 Employees by Attendance ⓘ



Early vs. Late Logins

23 Early 25 Late

Paid Time Off

30 hrs Available 43.50 hrs Used

Overall Attendance Percentage ⓘ



Leave Taken

20 Sick 40 Casual

Breaks vs. Overtime

41.52 hrs Break 145 hrs Overtime

Today's Worked Hours ⓘ



Attendance Statistics ⓘ

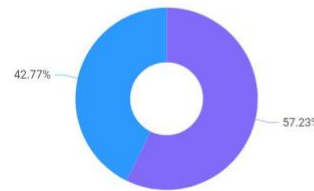
Last Month



Working Location ⓘ

Drill down status to see employees by shift.

Working from Home Working from Office

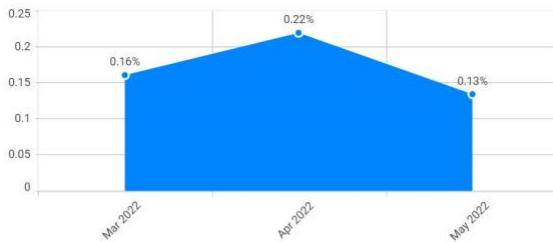


Current Month's Worked Hours ⓘ



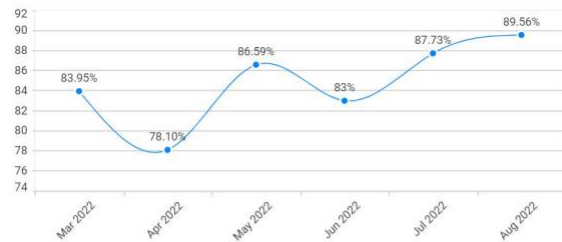
Absenteeism Rate by Month ⓘ

Last 6 Months

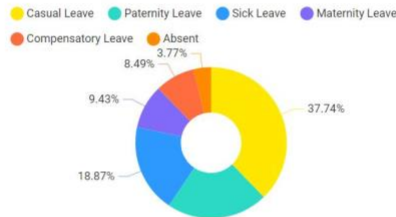


Overall Attendance by Month ⓘ

Last 6 Months



Leave Type Distribution ⓘ

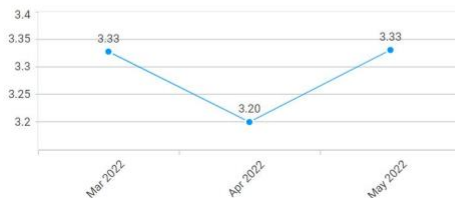


Upcoming Pre-Approved Absences ⓘ

Employee ID	Employee N...	Designation	Team	Leave Task ID	Date
S100	Alexander F...	Software En...	Data Source	LEAVESYS-9...	9/18/2022
S135	Daniel Tonini	Senior Cons...	Consulting s...	LEAVESYS-3...	9/16/2022

Performance Rating by Month ⓘ

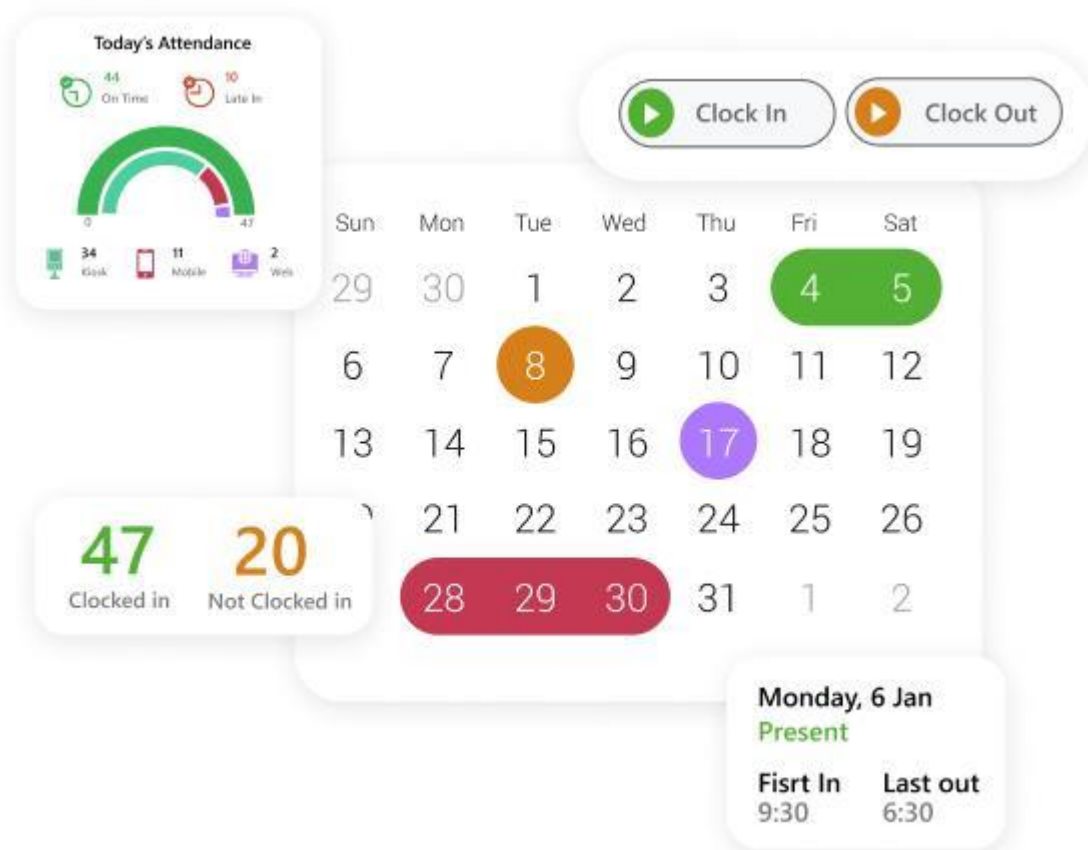
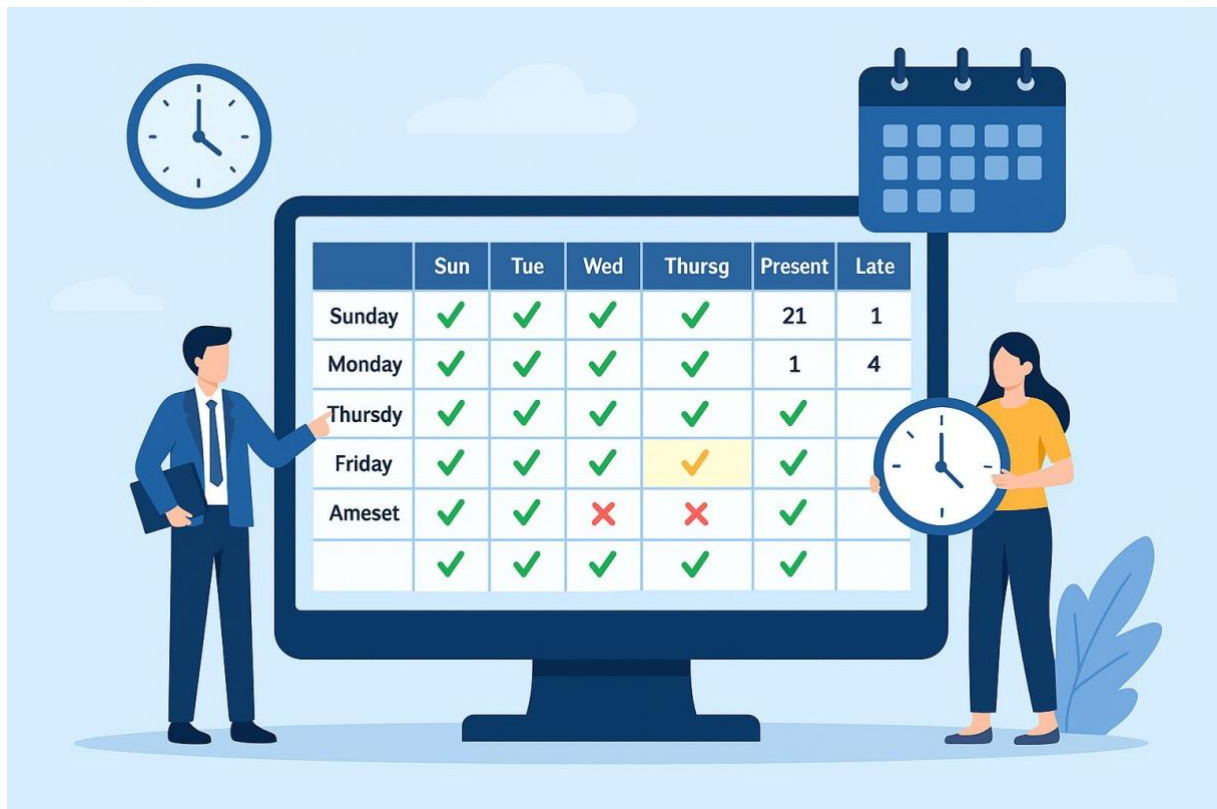
Last 6 Months

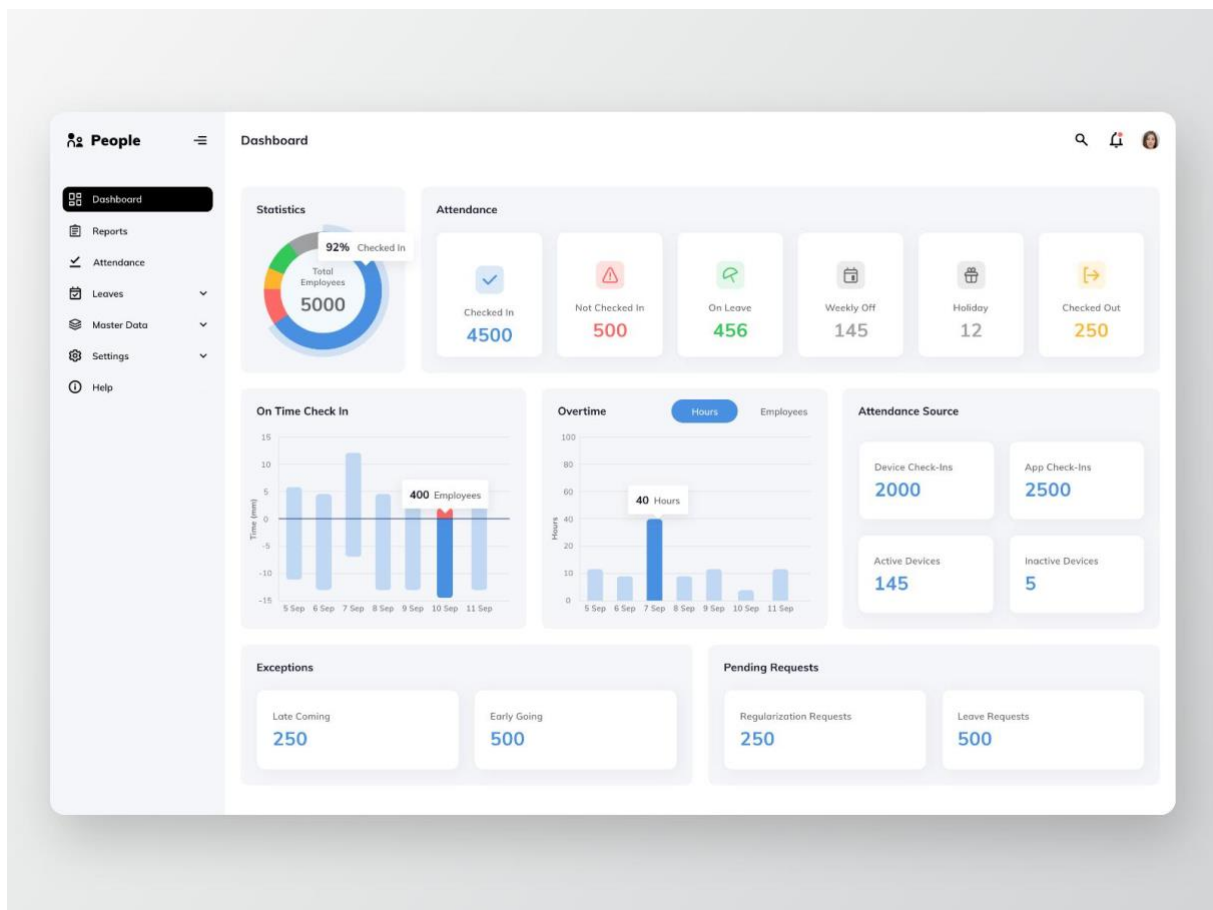
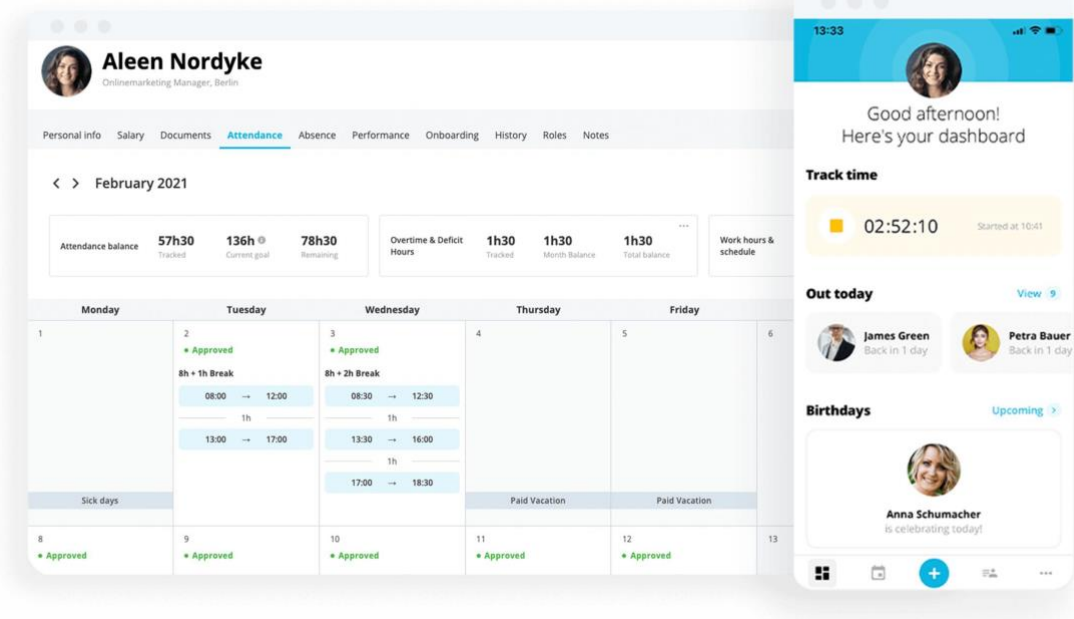


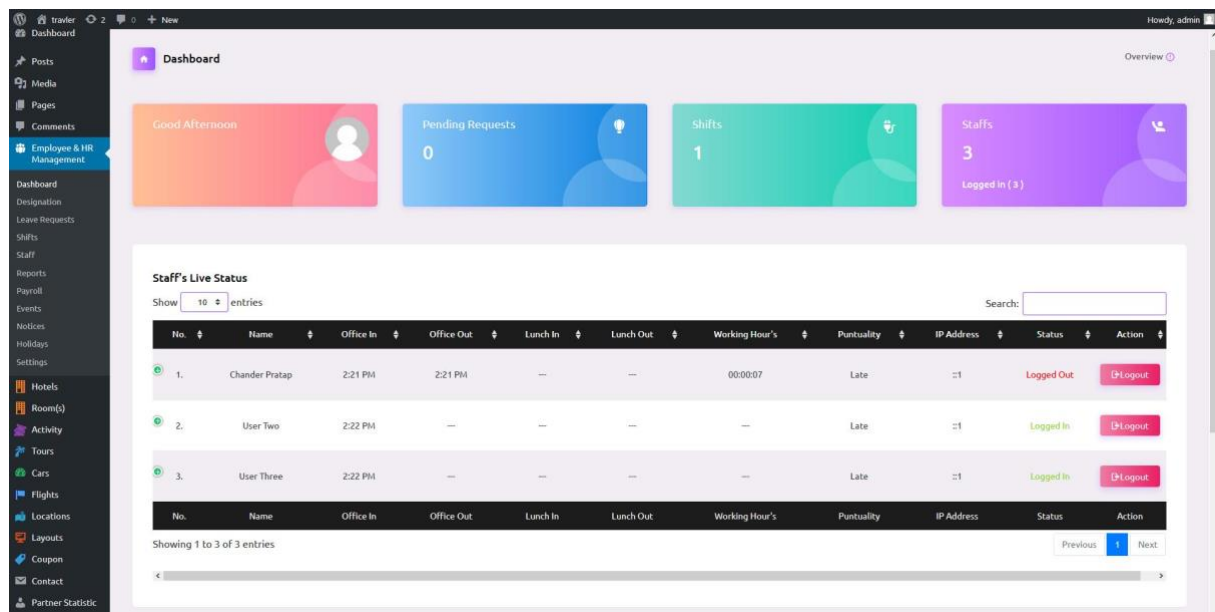
Employee Details ⓘ

Last 30 days

Employee ID	Employee N...	Date	Location	Shift	Login Time	Logout Time
S100	Alexander F...	8/15/2022	Working fro...	Day Shift	8:30:00 AM	6:10:00 PM
S100	Alexander F...	8/16/2022	Working fro...	Day Shift	9:30:00 AM	7:30:00 PM
S100	Alexander F...	8/17/2022	Working fro...	Day Shift	9:25:00 AM	6:00:00 PM







Indicateurs :

- Heures travaillées
- Retards
- Absences
- Congés
- Taux de présence

4.5 Filtres avancés

- Par jour
- Par mois
- Par année
- Par employé

4.6 Détection des anomalies

Manquements :

- Retard
- Absence
- Sortie anticipée

Insuffisance :

- Heures < seuil (ex : 8h/jour)

5. Modélisation MongoDB (Collections)

users

```
{
  "_id": "...",
  "nom": "Ali",
  "email": "ali@mail.com",
  "password": "...",
  "role": "admin | employe"
}
```

pointages

```
{
  "_id": "...",
  "user_id": "...",
  "date": "2026-04-22",
  "heure": "08:30",
  "type": "entree | sortie",
  "latitude": "...",
  "longitude": "...",
  "valide": true
}
```

conges

```
{
  "_id": "...",
  "user_id": "...",
  "date_debut": "...",
  "date_fin": "...",
  "type": "annuel",
  "statut": "en_attente | valide | refuse"
}
```


6. API (exemples)

Auth

- POST /login
- POST /register

Pointage

- POST /pointage
- GET /pointage?date=...

Congés

- POST /conge
- PUT /conge/:id

Dashboard

- GET /stats

7. Interfaces (Frontend React ou Next)

Employé

- Login
- Scanner QR
- Bouton pointer
- Historique personnel
- Demande de congé

Admin

- Dashboard
- Liste employés
- Gestion congés
- Rapport pointage
- Filtres

8. Sécurité

- JWT Authentication
- Vérification GPS
- QR code dynamique
- Validation backend obligatoire
- Protection API

9. Contraintes techniques

- Responsive design (mobile obligatoire)
- Temps réel (option WebSocket)
- Performance API
- Données sécurisées

10. Évolutions possibles

- Application mobile (React Native)
- Reconnaissance faciale
- Notifications
- Export Excel/PDF
- Intégration badge NFC

11. Livrables attendus

- Code source (Frontend + Backend) sur GitHub
- Base de données MongoDB
- Documentation technique
- Diagrammes (UML recommandé)
- Démo fonctionnelle
- Rapport + Présentation